Growth, Loss and Numerical Evidence

A Controlled Modeling Rehearsal

Codex MCM Adapter Demonstration

Abstract. We compare fourth-order Runge–Kutta (RK4) with a closed-form reference for a synthetic biomass model. At time 30, baseline biomass is 77.015951 units. The largest sampled baseline error is 2.99×10^{-8} units. Nine prescribed growth and loss combinations produce terminal biomass from 55.426868 to 90.469567 units. The resulting envelope describes the range over those settings. The rehearsal uses synthetic inputs throughout.

1 Problem Restatement and Analysis

The rehearsal asks whether RK4 matches a known trajectory, how growth and loss affect terminal biomass, and how far the trajectories diverge. Figure 1 traces the inputs and verification steps.

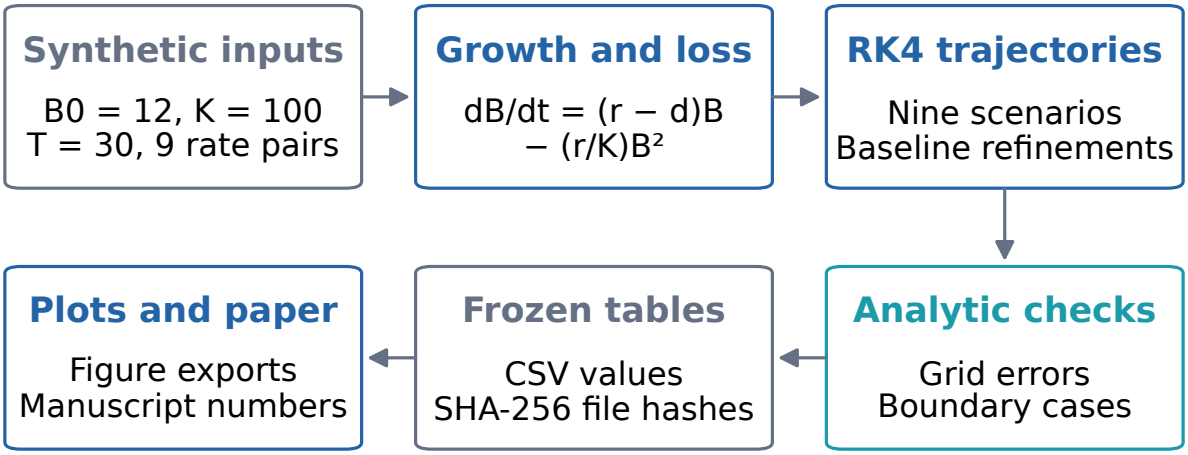


Figure 1: Evidence flow for the synthetic rehearsal. RK4 and the analytic reference share the model contract. CSVs supply plots and paper numbers. The data manifest records their SHA-256 hashes.

2 Assumptions and Notation

The system is spatially homogeneous with constant coefficients and nonnegative biomass. Table 1 defines the baseline and the units used in the synthetic calculations.

Table 1: Baseline quantities and units from the synthetic model contract.

Symbol	Meaning	Baseline value
B_0	Initial biomass	12 biomass units
r	Intrinsic growth rate	0.35 per time unit
d	Constant per-capita loss rate	0.08 per time unit
K	Growth carrying-capacity parameter	100 biomass units
T	Time horizon	30 time units
h	Integration step	0.1 time units

3 Model Construction and Solution

The balance between density-limited growth and loss gives

$$\dot{B} = rB\left(1 - \frac{B}{K}\right) - dB = aB - qB^2, \qquad a = r - d, \quad q = r/K. \tag{1}$$

For $a \neq 0$ and $B_0 > 0$, substituting $u = 1/B$ gives $\dot{u} + au = q$ and hence

$$B(t) = \frac{B_0 e^{at}}{1 + (qB_0/a)(e^{at} - 1)}. \tag{2}$$

At $a = 0$, integration yields $B(t) = B_0/(1 + qB_0 t)$. With $B_0 = 0$, biomass stays zero. RK4 uses a shortened final step when the horizon falls between grid points.

The comparison uses $E_h = \max_i |B_h(t_i) - B_{\text{exact}}(t_i)|$ on the solver grid. At $h = 0.1$, the error is 2.99×10^{-8} biomass units, below the contract's 10^{-3} threshold. Halving h from 0.2 to 0.1 and again to 0.05 reduces the error by factors of 15.97 and 15.98, close to the fourth-order factor of 16. Figure 2 shows where the numerical and analytic solutions differ over time.

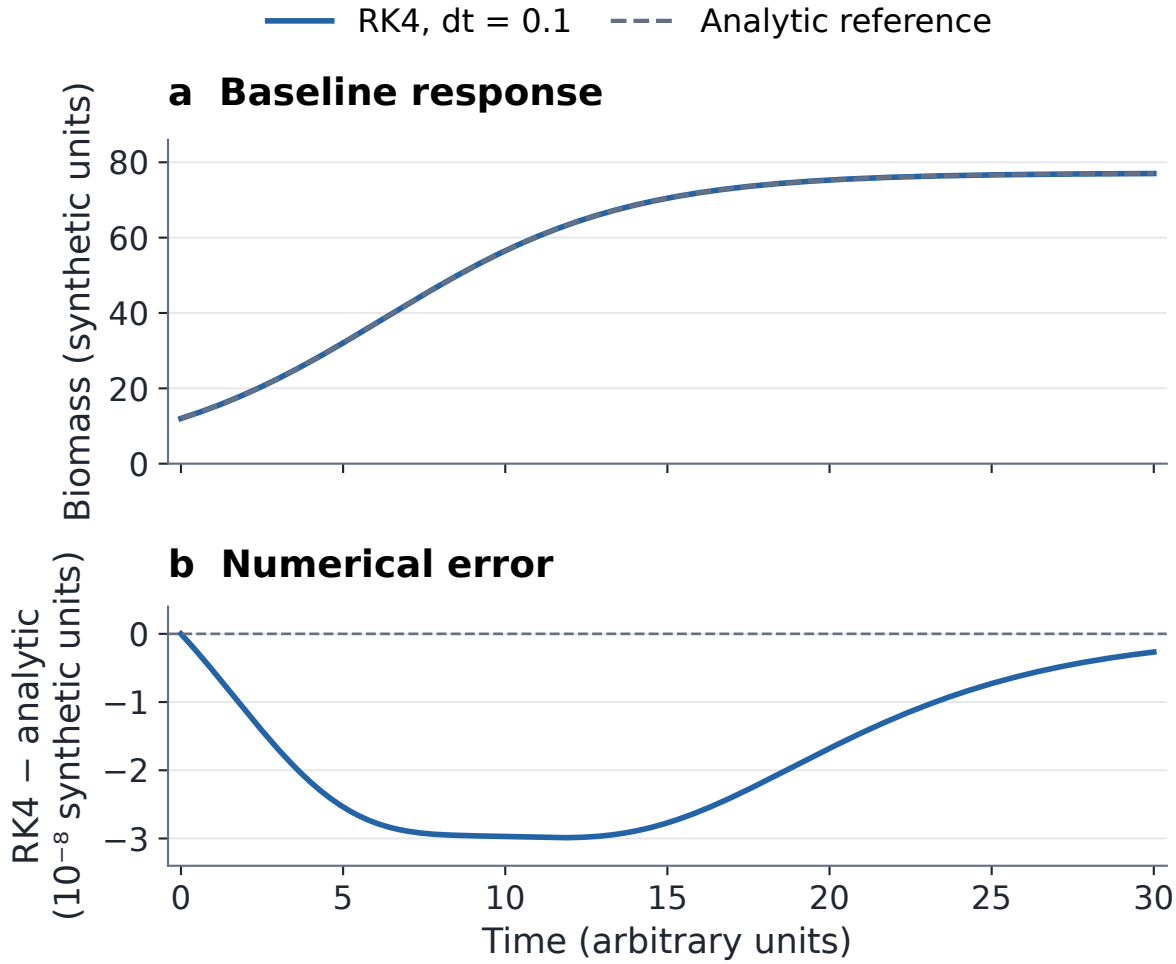


Figure 2: Baseline numerical verification at all 301 points with $h = 0.1$. (a) The RK4 and analytic biomass trajectories overlap at the displayed scale. (b) The signed difference is shown in units of 10^{-8} biomass.

4 Sensitivity Analysis

We vary $r \in \{0.28, 0.35, 0.42\}$ and $d \in \{0.04, 0.08, 0.12\}$, keeping K, B_0 and the horizon fixed. Figure 3 shows that terminal biomass increases with growth for each loss rate and decreases with loss for each growth rate. Across the nine settings, it ranges from 55.426868 to 90.469567 units.

For $r > d$, the positive equilibrium and its local derivatives are

$$B^* = K\left(1 - \frac{d}{r}\right), \tag{3}$$

$$\frac{\partial B^*}{\partial d} = -\frac{K}{r}, \qquad \frac{\partial B^*}{\partial r} = \frac{Kd}{r^2}. \tag{4}$$

These equilibrium derivatives agree with the directions observed at the finite horizon. The finite-time values depend on the initial state and the time available to approach equilibrium.

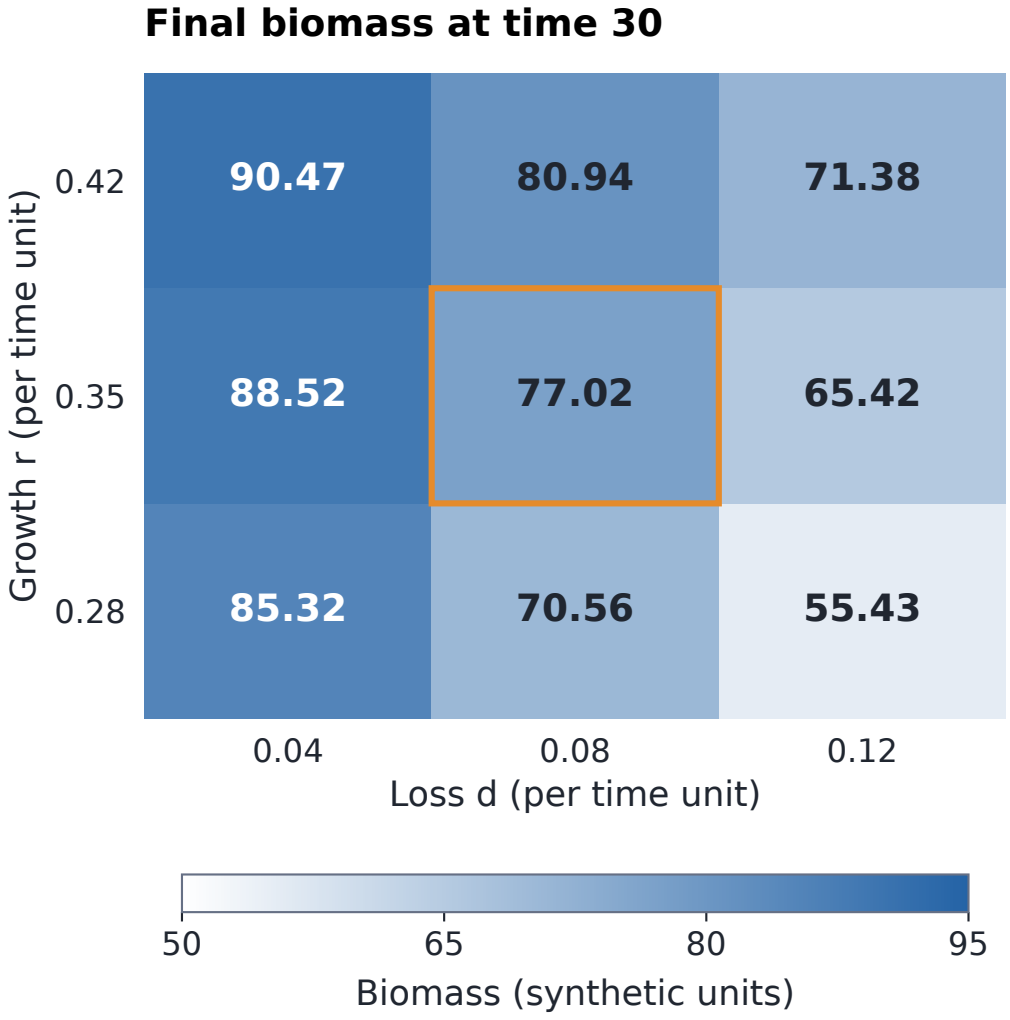


Figure 3: Terminal biomass at $T = 30$ across the nine tested growth and loss combinations. Each cell contains one RK4 result rounded to two decimals. The outlined center cell is the baseline.

5 Model Evaluation and Recommendations

The analytic comparison verifies RK4 accuracy. The parameter envelope uses the pointwise extrema of the nine trajectories in the prescribed set $\mathcal{S}$.

$$[L(t), U(t)] = \left[\min_{s\in\mathcal{S}} B_s(t),\ \max_{s\in\mathcal{S}} B_s(t)\right], \quad |\mathcal{S}| = 9. \tag{5}$$

Figure 4 shows the spread developing from a shared initial value. The bounds apply to the nine runs. Exceedance probabilities would require a distribution over inputs or empirical observations.

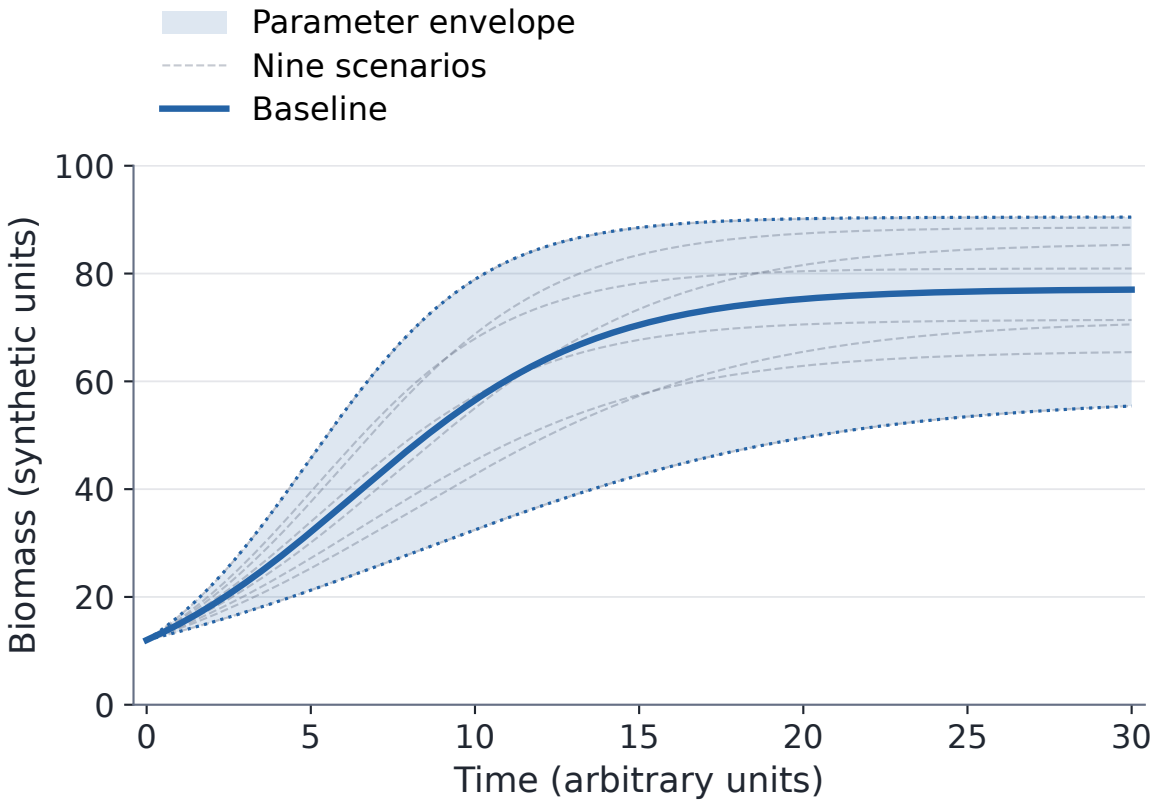


Figure 4: Pointwise minimum and maximum of nine trajectories on the common 301-point grid. Dashed lines show individual scenarios and the solid line marks the baseline.

Before interpreting the scenario results, verify numerical accuracy and report the parameter grid. An empirical application would also need calibration and an assessment of spatial effects.

Report on Use of AI Tools

Codex assisted with the model contract, code, figures, manuscript and verification.

A Reproduction

The example README provides the test and TeX build commands. Frozen CSVs, file hashes, and figure provenance accompany the manuscript in the evidence directory.